

**A PROJECT TITLE
ON
HEAVY VEHICLE FAILURE PREDICTION THROUGH
INTELLIGENT SENSOR DATA ANALYSIS**

Project report submitted in partial fulfilment of the requirements for the Course
MACHINE LEARNING

BACHELOR OF TECHNOLOGY
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1. ABSTRACT

Heavy vehicles operate under demanding conditions, and unexpected component failures can cause vehicle downtime, maintenance delays, increased operating costs, and safety risks. Traditional maintenance practices often depend on fixed service schedules and may fail to identify a developing fault before it becomes serious. This project proposes a machine learning based approach for predicting Air Pressure System (APS) component failures in heavy Scania trucks using sensor and operational data. The APS is an important truck system because pressurized air is used for functions such as braking and gear changes. The study uses the APS Failure at Scania Trucks dataset, which contains operational data collected from heavy Scania trucks in everyday usage. The combined dataset contains 76,000 records and 171 attributes, including anonymized operational measurements and histogram-based variables. The target variable represents whether the observed failure is related to the specified APS component. The dataset contains missing values and a highly imbalanced class distribution, making preprocessing an important part of the proposed system. The planned preprocessing includes handling missing values, analyzing feature distributions, identifying redundant or less informative attributes, encoding the target variable, and applying suitable techniques for class imbalance. Exploratory data analysis will be performed to understand relationships between operational sensor variables and APS failures. Machine learning classification algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost will be evaluated. Performance will be compared using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix, with particular attention to recall because missing an actual failure can be more costly than an unnecessary inspection. The objective is to develop a reliable failure prediction model that can provide early indications of APS-related faults and support proactive maintenance planning. The proposed system aims to reduce unexpected breakdowns, improve vehicle reliability, and demonstrate the practical application of machine learning to heavy-vehicle sensor intelligence.

2. INTRODUCTION

Heavy-duty trucks are essential for transportation and logistics, and their reliable operation is important for maintaining efficient fleet activities. Failure of critical vehicle components can result in unplanned downtime, maintenance expenses, and possible operational and safety problems. The Air Pressure System (APS) is particularly important because it generates pressurized air used in functions including braking and gear changes. Sensor and operational data collected during everyday truck usage can provide useful information about the condition of vehicle components. Machine learning can analyze such high-dimensional operational data and learn patterns associated with component failures. This project therefore focuses on predicting APS-related failures in Scania heavy trucks using machine learning. The study will combine the available training and test records into a single dataset for initial exploratory analysis, followed by appropriate preprocessing and a new data split for model development. Different classification algorithms will be compared to identify an effective predictive approach.

3. SOFTWARE REQUIREMENTS

Requirement	Technology
Programming Language	Python
Development Environment	Google Colab / Jupyter Notebook
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
ML Models	Logistic Regression, Decision Tree, Random Forest, SVM, XGBoost
Dataset	APS Failure at Scania Trucks
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

Signature of the Guide: _____

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